Homework 10

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

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Contents

Problem 7.2

Let X and Y be independent Bernoulli random variables with parameters p and r, respectively. Find the distribution of $X + Y$.

Solution.

Scratch / Setup. Let
$$S = X + Y.$$
Since $X, Y \in \{0, 1\}$, the only possible values of S are $0, 1, 2$.

The relevant combinations are:
$$\begin{aligned} S = 0 &\iff (X, Y) = (0, 0), \\ S = 1 &\iff (X, Y) \in \{(1, 0), (0, 1)\}, \\ S = 2 &\iff (X, Y) = (1, 1). \end{aligned}$$
Because X and Y are independent Bernoulli random variables,
$$P(X = 1) = p, \quad P(X = 0) = 1 - p, \quad P(Y = 1) = r, \quad P(Y = 0) = 1 - r.$$

We are computing the probability mass function of $S = X + Y$.

Since S can take only the values $0, 1, 2$, we compute these probabilities directly. First,
$$P(S = 0) = P(X = 0, Y = 0) = P(X = 0)P(Y = 0) = (1 - p)(1 - r).$$
Next,
$$P(S = 2) = P(X = 1, Y = 1) = P(X = 1)P(Y = 1) = pr.$$
Finally,
$$\begin{aligned} P(S = 1) &= P(X = 1, Y = 0) + P(X = 0, Y = 1) \\ &= P(X = 1)P(Y = 0) + P(X = 0)P(Y = 1) \\ &= p(1 - r) + (1 - p)r \\ &= p + r - 2pr. \end{aligned}$$

Therefore the pmf of $X + Y$ is
$$p_S(s) = P(S = s) = \begin{cases} (1 - p)(1 - r), & s = 0, \\ p + r - 2pr, & s = 1, \\ pr, & s = 2, \\ 0, & \text{otherwise.} \end{cases}$$

This is a binomial distribution only in the special case $p = r$.

Final: $\boxed{P(X + Y = 0) = (1 - p)(1 - r), \quad P(X + Y = 1) = p + r - 2pr}$

Final: $\boxed{P(X + Y = 2) = pr}$

Problem 7.16

Let X be a Poisson random variable with parameter λ, and Y an independent Bernoulli random variable with parameter p. Find the probability mass function of $X+Y$.

Solution.

Scratch / Setup. Let

$$S = X + Y.$$

Since $X \in \{0,1,2,\dots\}$ and $Y \in \{0,1\}$, the support of S is also

$$\{0,1,2,\dots\}.$$

The cleanest computation is to condition on Y:

$$P(S=n) = P(Y=0)P(X=n) + P(Y=1)P(X=n-1),$$

with the convention that $P(X=-1)=0$.

Also,

$$P(X=k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0,1,2,\dots$$

and

$$P(Y=0) = 1-p, \qquad P(Y=1) = p.$$

We want the probability mass function of $S = X+Y$.

For $n \geq 0$,

$$P(S=n) = P(Y=0)P(X=n) + P(Y=1)P(X=n-1).$$

If $n=0$, the second term is zero, so

$$P(S=0) = (1-p)P(X=0) = (1-p)e^{-\lambda}.$$

If $n \geq 1$, then

$$\begin{aligned} P(S=n) &= (1-p)e^{-\lambda}\frac{\lambda^n}{n!} + p\,e^{-\lambda}\frac{\lambda^{n-1}}{(n-1)!} \\ &= e^{-\lambda}\frac{\lambda^{n-1}}{n!}\big((1-p)\lambda + np\big). \end{aligned}$$

Hence the pmf of $X+Y$ is

$$p_{X+Y}(n) = \begin{cases} (1-p)e^{-\lambda}, & n = 0, \\ e^{-\lambda}\dfrac{\lambda^{n-1}}{n!}\big((1-p)\lambda + np\big), & n = 1,2,3,\dots \end{cases}$$

Final: $\boxed{p_{X+Y}(0) = (1-p)e^{-\lambda}, \quad p_{X+Y}(n) = e^{-\lambda}\frac{\lambda^{n-1}}{n!}\big((1-p)\lambda + np\big) \text{ for } n \geq 1}$

Problem 7.20

Let X have density $f_X(x) = 2x$ for $0 < x < 1$ and 0 outside of $(0, 1)$, and let Y be uniform on the interval $(1, 2)$. Assume X and Y independent.

(b) Find the density function of $X + Y$.

Solution.

Scratch / Setup. Let
$$T = X + Y.$$
Since $X \in (0, 1)$ and $Y \in (1, 2)$, we must have
$$T \in (1, 3).$$
By convolution,
$$f_T(t) = \int_{-\infty}^{\infty} f_X(x) f_Y(t - x)\, dx.$$
Here
$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise}, \end{cases} \qquad f_Y(y) = \begin{cases} 1, & 1 < y < 2, \\ 0, & \text{otherwise}. \end{cases}$$
The integrand is nonzero exactly when
$$0 < x < 1 \quad \text{and} \quad 1 < t - x < 2,$$
that is,
$$\max(0, t - 2) < x < \min(1, t - 1).$$
This leads to two cases:
$$1 < t < 2 \quad \text{and} \quad 2 \le t < 3.$$

We are finding the density of $T = X + Y$.

For any real t,
$$f_T(t) = \int_{-\infty}^{\infty} f_X(x) f_Y(t - x)\, dx.$$
If $t \notin (1, 3)$, there is no overlap of the supports, so $f_T(t) = 0$.

If $1 < t < 2$, then $\max(0, t - 2) = 0$ and $\min(1, t - 1) = t - 1$, so
$$f_T(t) = \int_0^{t-1} 2x\, dx = (t - 1)^2.$$
If $2 \le t < 3$, then $\max(0, t - 2) = t - 2$ and $\min(1, t - 1) = 1$, so
$$f_T(t) = \int_{t-2}^{1} 2x\, dx = 1 - (t - 2)^2.$$

Therefore

$$f_{X+Y}(t) = \begin{cases} (t-1)^2, & 1 < t < 2, \\ 1-(t-2)^2, & 2 \le t < 3, \\ 0, & \text{otherwise.} \end{cases}$$

As a quick check,

$$\int_1^2 (t-1)^2\,dt + \int_2^3 \bigl(1-(t-2)^2\bigr)\,dt = \frac{1}{3} + \frac{2}{3} = 1,$$

so this is a valid density.

Final: $\boxed{f_{X+Y}(t) = \begin{cases} (t-1)^2, & 1 < t < 2 \\ 1-(t-2)^2, & 2 \le t < 3 \\ 0, & \text{otherwise} \end{cases}}$

Problem 7.24

Suppose that X, Y, and Z are independent mean zero normal random variables. Find $P(X + 2Y - 3Z > 0)$.

Solution.

Scratch / Setup. Let

$$W = X + 2Y - 3Z.$$

Because X, Y, Z are independent normal random variables, any linear combination of them is also normal. So W is normal.

Its mean is

$$E[W] = E[X] + 2E[Y] - 3E[Z] = 0.$$

If we write

$$\mathrm{Var}(X) = \sigma_X^2, \quad \mathrm{Var}(Y) = \sigma_Y^2, \quad \mathrm{Var}(Z) = \sigma_Z^2,$$

then independence gives

$$\mathrm{Var}(W) = \sigma_X^2 + 4\sigma_Y^2 + 9\sigma_Z^2.$$

So W is a mean-zero normal random variable, hence its distribution is symmetric about 0.

We are asked to compute the probability

$$P(X + 2Y - 3Z > 0) = P(W > 0),$$

where

$$W = X + 2Y - 3Z.$$

Since W is a linear combination of independent normal random variables, it is itself normal. Its mean is

$$E[W] = 0,$$

and its variance is

$$\mathrm{Var}(W) = \mathrm{Var}(X) + 4\,\mathrm{Var}(Y) + 9\,\mathrm{Var}(Z).$$

Therefore W is a centered normal random variable. A mean-zero normal distribution is symmetric about 0, so

$$P(W > 0) = \frac{1}{2}.$$

Final: $\boxed{P(X + 2Y - 3Z > 0) = \frac{1}{2}}$

Problem 7.28

Let X_1, X_2, X_3 be independent Exp(λ) distributed random variables. Find the probability that $X_1 < X_2 < X_3$.

Solution.

Scratch / Setup. The variables X_1, X_2, X_3 are i.i.d. and continuous.

Because they are continuous,

$$P(X_i = X_j) = 0 \qquad \text{for } i \neq j,$$

so ties occur with probability 0.

There are $3! = 6$ strict orderings:

$$\begin{array}{lll} X_1 < X_2 < X_3, & X_1 < X_3 < X_2, & X_2 < X_1 < X_3, \\ X_2 < X_3 < X_1, & X_3 < X_1 < X_2, & X_3 < X_2 < X_1. \end{array}$$

By exchangeability of i.i.d. random variables, these six orderings are equally likely.

Since X_1, X_2, X_3 are independent and identically distributed continuous random variables, each strict ordering of the three variables has the same probability. Also, because the distribution is continuous, ties have probability zero. Hence the six events

$$\begin{array}{lll} X_1 < X_2 < X_3, & X_1 < X_3 < X_2, & X_2 < X_1 < X_3, \\ X_2 < X_3 < X_1, & X_3 < X_1 < X_2, & X_3 < X_2 < X_1 \end{array}$$

are disjoint, equally likely, and their union has probability 1.

Therefore each one has probability

$$\frac{1}{6}.$$

Final: $\boxed{P(X_1 < X_2 < X_3) = \frac{1}{6}}$

Problem 7.30

We deal five cards, one by one, from a standard deck of 52. (Dealing cards from a deck means sampling without replacement.)

(a) Find the probability that the second card is an ace and the fourth card is a king.

(b) Find the probability that the first and the fifth cards are both spades.

(c) Find the conditional probability that the second card is a king given that the last two cards are both aces.

Solution.

Scratch / Setup. We deal five cards from a uniformly shuffled deck, so all ordered 5-card sequences are equally likely.

Useful facts:

- There are 4 aces, 4 kings, and 13 spades.
- By symmetry/exchangeability, specified positions behave like any other specified positions.
- Conditioning on the last two cards being aces leaves 50 cards for the first three positions, still including all 4 kings.

We compute each part separately.

(a) We need the probability that the second card is an ace and the fourth card is a king.

By exchangeability of the positions in a random ordering of the deck, this is the same as the probability that the first card is an ace and the second card is a king. Therefore

$$P(\text{2nd ace and 4th king}) = \frac{4}{52} \cdot \frac{4}{51} = \frac{16}{2652} = \frac{4}{663}.$$

(b) We need the probability that the first and fifth cards are both spades.

Again by exchangeability, this is the same as the probability that the first two cards are both spades. Thus

$$P(\text{1st and 5th spades}) = \frac{13}{52} \cdot \frac{12}{51} = \frac{1}{17}.$$

(c) We need

$$P(\text{2nd card is a king} \mid \text{last two cards are both aces}).$$

After conditioning on the last two cards being aces, the first three positions are filled by an ordered sample from the remaining 50 cards. None of the aces is a king, so all 4 kings are still among those 50 cards. By symmetry, each of the first three positions is equally likely to contain any one of the remaining 50 cards. Hence

$$P(\text{2nd card is a king} \mid \text{last two cards are both aces}) = \frac{4}{50} = \frac{2}{25}.$$

Final: $\boxed{\text{(a) } \frac{4}{663}}$

Final: $\boxed{\text{(b) } \frac{1}{17}}$

Final: $\boxed{\text{(c) } \frac{2}{25}}$